Started on	Saturday, 10 October 2026, 10:41 PM
State	Finished
Completed on	Saturday, 10 October 2026, 11:07 PM
Time taken	25 mins 36 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    if(scanf("%d", &n)!=1) return 0;
    int arr[n];
    for(int i=0;i<n;i++){
        while(scanf("%d", &arr[i])!=1){
            getchar();
        }

    }
    int dp[n];
    int max_len=0;
    for(int i=0;i<n;i++){
        dp[i]=1;
        for(int j=0;j<i;j++){
            if(arr[j]<=arr[i]){
                if(dp[j]+1>dp[i]){
                    dp[i]=dp[j]+1;
                }
        }
    }
    if (dp[i]>max_len){
        max_len=dp[i];
    }
}
    printf("%d\n",max_len);
    return 0;
}
```

	Input	Expected	Got	
✔	9 −1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.